Engineering Expo BSChE Project Team: Judge Score Assessment- Spring 2018

Scoring Rubrics Mapped to Student Learning Outcomes and Program Educational Objectives

RESULTS BY PERFORMANCE INDICATOR: Prelab, Final Lab Reports												
Rubric item (PI)	1	2	3	4	5	6	7	8	9	10	11	12
Mapped SLO	2	2	2	2	2	2	3	3	3	3	3	3
Average	2.95	2.63	2.76	2.83	2.61	2.58	2.88	2.83	2.90	2.75	2.81	2.78
Std Dev	0.21	0.49	0.43	0.36	0.49	0.50	0.33	0.38	0.29	0.50	0.45	0.42

RESULTS BY STUDENT OUTCOME ALL ASSESSMENTS	2	3
Average Score	2.73	2.82
Standard dev.	0.15	0.06

Technical Quality

Performance Indicator (PI)	**SLO**	**(1)** Unsatisfactory	**(2)** Developing	**(3)** Satisfactory
1. Identification of project objective(s)	**2**	No design strategy; haphazard approach	Uses a design strategy with guidance	Develops a design strategy, decomposition of work into subtasks
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	**2**	No consideration of economics, safety, and environment	Can use prior knowledge to design individual pieces of equipment competently when guided to do so	Understands how areas interrelate and demonstrates ability to integrate prior knowledge into a new problem
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	**2**	Cannot design processes or individual pieces of equipment without significant amounts of help	Can follow a previous example competently	Suggests new approaches and improves on what has been done before
4. Overall fulfillment of project objectives	**2**	No consideration of economics, safety, and environment	Design is done, but procedures and equations are not documented or referenced	Develops a solution that includes economic, safety, environmental and other realistic constraints
5. Project innovation	**2**	Unable to relate prior knowledge to the design problem	Can use prior knowledge to design individual pieces of equipment competently when guided to do so	Understands how areas interrelate and demonstrates ability to integrate prior knowledge into a new problem
6. Project practicality in view of contemporary engineering standards/practice	**2**	No application of engineering and/or scientific principles	Includes only minor or cursory consideration of economic, safety, and environmental constraints	Applies engineering and/or scientific principles correctly to design practical processes

Presentation Quality

Performance Indicator (PI)	**SLO**	**(1)** Unsatisfactory	**(2)** Developing	**(3)** Satisfactory
7. General professionalism and oral communication	**3**	Talk is poorly organized, e.g. no clear introduction or summary of talk is presented	Presents key elements of an oral presentation adequately, but "tell them" not clearly applied	Plans and delivers an oral presentation effectively; applies the principle of "(tell them)" --well organized
8. Knowledge of the background material	**3**	Unable to relate prior knowledge to the design problem	Can use prior knowledge to design individual pieces of equipment competently when guided to do so	Understands how areas interrelate and demonstrates ability to integrate prior knowledge into a new problem
9. Organization & quality of presentation material	**3**	Organizes materials in a logical sequence to enhance the reader's comprehension	Material organized well, but some sections and sub-sections are not identified clearly	Graphs, tables or diagrams are not easy to understand
10. Project description (see Program Book)	**3**	Text rambles, points made are only understood with repeated reading, and key points are not organized	Articulates ideas, but writing is somewhat disjointed, superfluous or difficult to follow	Articulates ideas clearly and concisely
11. Group dynamics/member participation	**3**	No interaction between team members. One member only talks	Interaction between team members, but one person leads talk	Discussion by all team members; all understand project
12. Quality, effectiveness, & use of visual aids	**3**	Multiple slides are unclear or incomprehensible. Presents well mechanically: · Makes eye contact · Can be easily heard · Speaks comfortably with minimal prompts (notecards) · No distracting nervous habits	Visual aides have minor errors or are not always clearly visible. Has some minor difficulties with the mechanical aspects of the presentation · Eye contact is sporadic · Occasionally difficult to hear or understand speaking	Uses visual aides effectively. Presents well mechanically: · Makes eye contact · Can be easily heard · Speaks comfortably with minimal prompts (notecards) · No distracting nervous habits